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Entangling Unravellings for Quantum State Engineering and Quantum Metrology

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A coloro che mi hanno insegnato a non mollare mai

True happiness is to enjoy the present, without anxious dependence upon the future, not to amuse ourselves with either hopes or fears but to rest satisfied with what we have, which is sufficient, for he that is so wants nothing. The greatest blessings of mankind are within us and within our reach. A wise man is content with his lot, whatever it may be, without wishing for what he has not.
- Seneca

Contents

1	Two qubit independent dephasing system: looking for entanglement	1
1.1	Convention, states and master equation	1
1.2	Unitary invariance of Lindblad operators	3
1.3	Photodetection	4
1.3.1	No-jump evolution of two-qubits	4
1.3.2	First jump from $ ++\rangle$	6
1.3.3	Subsequent jumps: two-state cyclicity	7
1.3.4	Dynamics of CSS states	8
1.3.5	Analyzing evolution and jumps: even-parity channel	8
1.3.6	Analyzing evolution and jumps: odd-parity channel case	11
1.3.7	Ensemble-averaged quantities over trajectories	13
1.3.8	Spin-squeezing	14
1.4	Homodyne detection	17
1.4.1	Pure-state unravelling: homodyne SSE	19

1

Two qubit independent dephasing system: looking for entanglement

1.1 Convention, states and master equation

We consider a system of $N = 2$ qubits, each described in the computational basis $\{|0\rangle, |1\rangle\}$. Throughout this work we adopt the convention

$$\sigma_z |1\rangle = +|1\rangle, \quad \sigma_z |0\rangle = -|0\rangle, \quad (1.1)$$

so that the excited state $|1\rangle$ corresponds to the positive eigenvalue of σ_z . This choice fixes all relative signs in the collective spin operators and in the dissipative dynamics. The single-qubit spin operators are defined as

$$S_\alpha^{(i)} = \frac{1}{2} \sigma_\alpha^{(i)}, \quad \alpha \in \{x, y, z\}, \quad (1.2)$$

where the superscript i labels the qubit. Collective spin operators are introduced as

$$J_\alpha = \sum_{i=1}^N S_\alpha^{(i)}. \quad (1.3)$$

When needed, spin coherent states (CSS) are written as

$$|\theta, \phi\rangle_N = \bigotimes_{i=1}^N \left(e^{i\phi} \sin \frac{\theta}{2} |1\rangle_i + \cos \frac{\theta}{2} |0\rangle_i \right), \quad (1.4)$$

which corresponds to a product state with all spins pointing along the direction $\hat{n} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$ on the Bloch sphere. We explicit the case for $N=2$ and $\phi = 0$:

$$|\theta\rangle \equiv |\theta, 0\rangle_2 = \sin^2 \frac{\theta}{2} |11\rangle + \sin \frac{\theta}{2} \cos \frac{\theta}{2} (|10\rangle + |01\rangle) + \cos^2 \frac{\theta}{2} |00\rangle. \quad (1.5)$$

In particular, the eigenstates of σ_x are

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}. \quad (1.6)$$

We use the standard Bell states

$$|\phi_\pm\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}} \quad (\text{even } z\text{-parity}), \quad (1.7)$$

$$|\psi_\pm\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}} \quad (\text{odd } z\text{-parity}). \quad (1.8)$$

The dynamics of the system is described within the Markovian open-quantum-system framework. Under the standard assumptions of weak system-environment coupling, short bath correlation time (Born-Markov approximation), and complete positivity, the reduced density operator $\rho(t)$ obeys a Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) master equation,

$$\frac{d\rho}{dt} = \mathcal{L}_k[\rho_t] = -i[H, \rho^{(c)}] + \sum_k \mathcal{D}[C_k]\rho^{(c)} dt = -i[H, \rho] + \sum_k \left(C_k \rho C_k^\dagger - \frac{1}{2} \{C_k^\dagger C_k, \rho\} \right). \quad (1.9)$$

where $\mathcal{D}[A]\rho = A\rho A^\dagger - \frac{1}{2}\{A^\dagger A, \rho\}$ is called the dissipator superoperator. Here H is the system Hamiltonian, and $\{C_k\}$ are the Lindblad or collapse operators describing dissipative processes and continuous measurements. We remark that equation (1.9) generates a completely positive and trace-preserving (CPTP) dynamical semigroup.

In this work we focus on dissipative channels associated with spin components along the z direction that acts independently on each qubit with strength γ . We can therefore write the master equation as:

$$\frac{d\rho_t}{dt} = -i[H, \rho_t] + \sum_{k=1}^2 \mathcal{D}[\sigma_z^{(k)}]\rho_t^{(c)} dt \quad (1.10)$$

By exploiting $\sigma_z^\dagger = \sigma_z$ and $\sigma_z^2 = \mathbb{1}$ we can rewrite (1.10) as:

$$\frac{d\rho_t}{dt} = \mathcal{L}_\gamma = -i\frac{\omega}{2} \sum_{k=1}^2 [\sigma_z^{(k)}, \rho_t] + \frac{\gamma}{2} \left(\sum_{k=1}^2 \sigma_z^{(k)} \rho_t \sigma_z^{(k)} - N \rho_t \right) \quad (1.11)$$

A central role is played by collective jump operators of the form:

$$\hat{C}_\pm = \sqrt{\gamma} \frac{\sigma_z^{(1)} \pm \sigma_z^{(2)}}{\sqrt{2}} \quad (1.12)$$

These operators naturally arise in measurement schemes where emissions from the two qubits are interfered (e.g. via a beam splitter) before detection, such that the measurement does not reveal which-path information. The two channels correspond to symmetric and antisymmetric combinations of local z -spin operators. While the master equation (1.9) uniquely determines the evolution of the density operator, its stochastic unravelling into quantum trajectories is not unique (see Appendix A). Different measurement schemes (photodetection, homodyne, heterodyne) correspond to different stochastic evolutions at the level of pure states, all reproducing the same ensemble-averaged dynamics.

In the photodetection unravelling associated with the jump operators (1.12), the system undergoes non-unitary no-click evolution interrupted by quantum jumps. Each jump corresponds to the application of $\hat{C}_\pm$ to the state vector, followed by normalization, and reveals information about collective spin observables and parity.

This measurement backaction is the key mechanism exploited in this work for the generation and stabilization of entanglement and for the study of quantities such as concurrence and spin squeezing.

1.2 Unitary invariance of Lindblad operators

Let U be an $n \times n$ unitary matrix acting on the vector of collapse operators $\{\hat{C}_i\}$. We prove the identity

$$\sum_{i=1}^n \mathcal{D}[\hat{C}_i] \rho = \sum_{i=1}^n \mathcal{D} \left[\sum_{j=1}^n U_{ij} \hat{C}_j \right] \rho, \quad (1.13)$$

where $\mathcal{D}[A] \rho = A \rho A^\dagger - \frac{1}{2} \{A^\dagger A, \rho\}$.

We expand the right-hand side:

$$\begin{aligned} \sum_i \mathcal{D} \left[\sum_j U_{ij} \hat{C}_j \right] \rho &= \sum_i \left[\left(\sum_j U_{ij} \hat{C}_j \right) \rho \left(\sum_k U_{ik} \hat{C}_k \right)^\dagger - \frac{1}{2} \left\{ \left(\sum_j U_{ij} \hat{C}_j \right)^\dagger \left(\sum_k U_{ik} \hat{C}_k \right), \rho \right\} \right] \\ &= \sum_i \left[\sum_{j,k} U_{ij} U_{ik}^* \hat{C}_j \rho \hat{C}_k^\dagger - \frac{1}{2} \left\{ \sum_{j,k} U_{ij}^* U_{ik} \hat{C}_j^\dagger \hat{C}_k, \rho \right\} \right]. \end{aligned} \quad (1.14)$$

We interchange the sums and group the coefficients:

$$\sum_i \mathcal{D} \left[\sum_j U_{ij} \hat{C}_j \right] \rho = \sum_{j,k} \left[\left(\sum_i U_{ij} U_{ik}^* \right) \hat{C}_j \rho \hat{C}_k^\dagger - \frac{1}{2} \left\{ \left(\sum_i U_{ij}^* U_{ik} \right) \hat{C}_j^\dagger \hat{C}_k, \rho \right\} \right]. \quad (1.15)$$

Since U is unitary, $U^\dagger U = \mathbb{1}$, hence

$$\sum_i U_{ij}^* U_{ik} = \delta_{jk}, \quad \sum_i U_{ij} U_{ik}^* = \delta_{jk}. \quad (1.16)$$

Substituting in Eq. (1.15) gives

$$\sum_i \mathcal{D} \left[\sum_j U_{ij} \hat{C}_j \right] \rho = \sum_j \left[\hat{C}_j \rho \hat{C}_j^\dagger - \frac{1}{2} \{ \hat{C}_j^\dagger \hat{C}_j, \rho \} \right] = \sum_j \mathcal{D}[\hat{C}_j] \rho. \quad (1.17)$$

This proves Eq. (1.13). ■

Applying this invariance to our system, the Lindblad master equation can be equivalently written as

$$\begin{aligned} d\rho &= -i[H, \rho] dt + \gamma \mathcal{D}[\sigma_z^{(1)}] \rho dt + \gamma \mathcal{D}[\sigma_z^{(2)}] \rho dt \\ &= -i[H, \rho] dt + \gamma \mathcal{D} \left[\frac{\sigma_z^{(1)} + \sigma_z^{(2)}}{\sqrt{2}} \right] \rho dt + \gamma \mathcal{D} \left[\frac{\sigma_z^{(1)} - \sigma_z^{(2)}}{\sqrt{2}} \right] \rho dt. \end{aligned} \quad (1.18)$$

The last equality follows by applying the Hadamard unitary

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (1.19)$$

to the operator vector $(\sigma_z^{(1)}, \sigma_z^{(2)})^T$.

1.3 Photodetection

A Markovian master equation admits a stochastic *unravelling* in terms of measurement-conditioned quantum trajectories. In the case of photodetection (quantum-jump monitoring), the measurement record is a set of counting processes $\{N_k(t)\}_k$ associated with the monitored output channels. In an infinitesimal time step dt , the increments

$$dN_k(t) \in \{0, 1\} \quad (1.20)$$

represent, respectively, a *no-click* event or a *click* in detector k . They are Poissonian random variables with conditional mean

$$\mathbb{E}[dN_k(t)] = \eta \text{Tr} \left[\hat{C}_k^\dagger \hat{C}_k \rho^{(c)}(t) \right] dt, \quad (1.21)$$

where $\eta \in [0, 1]$ is the detection efficiency and $\rho^{(c)}(t)$ denotes the conditional (normalized) system state.

For jump monitoring, the conditional evolution can be written as a piecewise deterministic process: a smooth non-unitary evolution conditioned on $dN_k = 0$, interrupted by stochastic jumps when $dN_k = 1$. A convenient Itô form of the stochastic master equation (SME) is

$$\begin{aligned} d\rho^{(c)} = & -i[H, \rho^{(c)}] dt + (1 - \eta) \sum_k \mathcal{D}[\hat{C}_k] \rho^{(c)} dt \\ & + \sum_k \left(\frac{\hat{C}_k \rho^{(c)} \hat{C}_k^\dagger}{\text{Tr}[\hat{C}_k^\dagger \hat{C}_k \rho^{(c)}]} - \rho^{(c)} \right) dN_k(t), \end{aligned} \quad (1.22)$$

By averaging Eq. (1.22) over the measurement outcomes one recovers the unconditional GKSL master equation.

The resulting photodetection SME reads

$$d\rho_t^{(c)} = -i\omega[J_z, \rho_t^{(c)}]dt + (1 - \eta) \frac{\gamma}{2} \sum_j \mathcal{D}[\sigma_z^{(j)}] \rho_t^{(c)} dt + \sum_j \left(\sigma_z^{(j)} \rho_t^{(c)} \sigma_z^{(j)} - \rho_t^{(c)} \right) dN_j, \quad (1.23)$$

with $\mathbb{E}[dN_j] = \eta\gamma dt$. A crucial remark (highly relevant for interpreting measurement records) is that, since $\sigma_z^{(j)}$ is unitary, the jump statistics is state-independent: the records $N_j(t)$ contain only noise and no information about the system state, and the SME is linear in $\rho_t^{(c)}$. This follows from the probability in measuring one photon:

$$p_1(t + dt) = \text{Tr}[\rho_1^{(c)}(t + dt)] = \gamma \cdot \langle \hat{C}^\dagger \hat{C} \rangle_t dt \quad (1.24)$$

We can see affirm that if $\hat{C}$ is unitary, it gives only $\gamma \cdot dt$. Since we have two different collapsing channels, we have to consider $2\gamma \cdot dt$

1.3.1 No-jump evolution of two-qubits

In this section we derive the no-jump evolution of a two-qubit under the z -dephasing dynamics introduced above. Since the meaning of no-jump depends on the detection efficiency, we treat separately the cases $\eta = 1$, perfect photodetection, and $\eta < 1$ (inefficient photodetection, i.e. “no detected clicks”).

For $\eta = 1$, no jump means that no click occurred in either monitored channel up to time t . In the quantum-jump picture, the unnormalized conditional pure state evolves under the non-Hermitian effective Hamiltonian

$$\hat{H}_{\text{eff}} = \hat{H} - \frac{i}{2} \sum_k \hat{C}_k^\dagger \hat{C}_k, \quad (1.25)$$

so that

$$|\tilde{\psi}(t)\rangle = e^{-i\hat{H}_{\text{eff}}t} |\psi(0)\rangle, \quad (1.26)$$

In our case (pure dephasing) we can work with the collective jump operators (1.12) and properties of σ_z to obtain

$$\hat{C}_+^\dagger \hat{C}_+ + \hat{C}_-^\dagger \hat{C}_- = \frac{\gamma}{2} \left[(\sigma_z^{(1)} + \sigma_z^{(2)})^2 + (\sigma_z^{(1)} - \sigma_z^{(2)})^2 \right] = 2\gamma \mathbb{1}. \quad (1.27)$$

For $\hat{H} = 0$, Eq. (1.25) gives $\hat{H}_{\text{eff}} = -i\gamma \mathbb{1}$ and therefore

$$|\tilde{\psi}(t)\rangle = e^{-\gamma t} |\psi(0)\rangle. \quad (1.28)$$

After normalization, the global factor cancels and the conditional state is unchanged:

$$|\psi(t)\rangle_{\text{no jump}} = |\psi(0)\rangle \quad (\eta = 1). \quad (1.29)$$

In particular, for any initial CSS one has

$$|\theta, \phi\rangle \xrightarrow{\text{no jump}} |\theta, \phi\rangle \quad (\eta = 1). \quad (1.30)$$

For $\eta < 1$, conditioning on “no detected clicks” does not imply that no quantum jumps occurred: unobserved emissions contribute an additional unconditional evolution term. The conditional dynamics in the absence of detected counts is deterministic and coincides with the unmonitored part of the SME. With our dephasing model, the no-detected-click evolution is governed by

$$\frac{d\rho(t)}{dt} = \mathcal{L}_{(1-\eta)\gamma}^{(2)}[\rho(t)] = \frac{(1-\eta)\gamma}{2} \sum_{j=1}^2 \left(\sigma_z^{(j)} \rho(t) \sigma_z^{(j)} - \rho(t) \right), \quad (1.31)$$

This is a local dephasing channel acting independently on each qubit. Since the initial CSS is a product state, the solution factorizes:

$$\rho_{\text{CSS},\theta}^{(\text{no click})}(t) = \rho_1^{(\text{no click})}(t) \otimes \rho_1^{(\text{no click})}(t), \quad (1.32)$$

where $\rho_1^{(\text{no click})}(t)$ is the single-qubit solution starting from the pure state $|\theta, \phi\rangle$.

Writing the initial single-qubit density operator in the basis $\{|0\rangle, |1\rangle\}$,

$$\rho_1(0) = |\theta, \phi\rangle \langle \theta, \phi| = \begin{pmatrix} \cos^2 \frac{\theta}{2} & e^{-i\phi} \sin \frac{\theta}{2} \cos \frac{\theta}{2} \\ e^{i\phi} \sin \frac{\theta}{2} \cos \frac{\theta}{2} & \sin^2 \frac{\theta}{2} \end{pmatrix}, \quad (1.33)$$

one finds that the populations are constant and the coherences decay exponentially:

$$\rho_1^{(\text{no click})}(t) = \begin{pmatrix} \cos^2 \frac{\theta}{2} & e^{-i\phi} \sin \frac{\theta}{2} \cos \frac{\theta}{2} e^{-(1-\eta)\gamma t} \\ e^{i\phi} \sin \frac{\theta}{2} \cos \frac{\theta}{2} e^{-(1-\eta)\gamma t} & \sin^2 \frac{\theta}{2} \end{pmatrix} \quad (\eta < 1). \quad (1.34)$$

Equations (1.32) and (1.34) provide the full two-qubit no-detected-click state for any (θ, ϕ) .

For later use, we report the explicit two-qubit matrix for $\phi = 0$ in the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$. Introducing the shorthand $c \equiv \cos \frac{\theta}{2}$ and $s \equiv \sin \frac{\theta}{2}$ one obtains

$$\rho_{\text{CSS},\theta}^{(\text{no click})}(t) = \begin{pmatrix} c^4 & c^3 s e^{-(1-\eta)\gamma t} & c^3 s e^{-(1-\eta)\gamma t} & c^2 s^2 e^{-2(1-\eta)\gamma t} \\ c^3 s e^{-(1-\eta)\gamma t} & c^2 s^2 & c^2 s^2 e^{-2(1-\eta)\gamma t} & c s^3 e^{-(1-\eta)\gamma t} \\ c^3 s e^{-(1-\eta)\gamma t} & c^2 s^2 e^{-2(1-\eta)\gamma t} & c^2 s^2 & c s^3 e^{-(1-\eta)\gamma t} \\ c^2 s^2 e^{-2(1-\eta)\gamma t} & c s^3 e^{-(1-\eta)\gamma t} & c s^3 e^{-(1-\eta)\gamma t} & s^4 \end{pmatrix}. \quad (1.35)$$

1.3.2 First jump from $|++\rangle$

We work in the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, and we adopt the convention $\sigma_z |0\rangle = -|0\rangle$, $\sigma_z |1\rangle = +|1\rangle$. The single-qubit $|+\rangle$ state is $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, hence

$$|++\rangle = |+\rangle \otimes |+\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) = \frac{1}{2}(1, 1, 1, 1)^T \quad (1.36)$$

We define the two-qubit z -parity operator as

$$\Pi_z \equiv \sigma_z^{(1)} \sigma_z^{(2)}. \quad (1.37)$$

It is Hermitian and unitary: $\Pi_z^\dagger = \Pi_z$ and $\Pi_z^2 = \mathbb{1}$, hence its eigenvalues are ± 1 . In the computational basis one has $\Pi_z |00\rangle = +|00\rangle$, $\Pi_z |11\rangle = +|11\rangle$, while $\Pi_z |01\rangle = -|01\rangle$, $\Pi_z |10\rangle = -|10\rangle$. Therefore its matrix representation is diagonal:

$$\Pi_z \equiv \begin{pmatrix} +1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & +1 \end{pmatrix}. \quad (1.38)$$

The projectors onto the even/odd parity subspaces are $\Pi_{\text{even}} = \frac{1}{2}(\mathbb{1} + \Pi_z)$ and $\Pi_{\text{odd}} = \frac{1}{2}(\mathbb{1} - \Pi_z)$. A useful property for what follows is that $|++\rangle$ (1.36) has equal weight on the even- and odd-parity computational sectors: the even sector is spanned by $\{|00\rangle, |11\rangle\}$, the odd one by $\{|01\rangle, |10\rangle\}$.

Since $\sigma_z^{(1)}$ and $\sigma_z^{(2)}$ commute and $(\sigma_z^{(i)})^2 = \mathbb{1}$,

$$[\sigma_z^{(1)}, \Pi_z] = \sigma_z^{(1)} \sigma_z^{(1)} \sigma_z^{(2)} - \sigma_z^{(1)} \sigma_z^{(1)} \sigma_z^{(2)} = 0, \quad (1.39)$$

$$[\sigma_z^{(2)}, \Pi_z] = \sigma_z^{(2)} \sigma_z^{(1)} \sigma_z^{(2)} - \sigma_z^{(1)} \sigma_z^{(2)} \sigma_z^{(2)} = \sigma_z^{(1)} - \sigma_z^{(1)} = 0. \quad (1.40)$$

Therefore $[\sigma_z^{(1)} \pm \sigma_z^{(2)}, \Pi_z] = 0$, and by (1.12) we have

$$[\hat{C}_\pm, \Pi_z] = 0. \quad (1.41)$$

This is a key result: a jump produced by $\hat{C}_\pm$ preserves the z -parity: if a pre-jump pure state $|\psi\rangle$ is a parity eigenstate, so is the post-jump normalized state : $|\psi'\rangle = \hat{C}_\pm |\psi\rangle / \|\hat{C}_\pm |\psi\rangle\|$. Let us work on the collapsing operators, by calculating:

$$\hat{C}_\pm^\dagger \hat{C}_\pm = (\mathbb{1} \pm \Pi_z). \quad (1.42)$$

we see that for pure-state jumps we use the standard update rule

$$p_\pm = \langle \psi | \hat{C}_\pm^\dagger \hat{C}_\pm | \psi \rangle dt = \langle \psi | \mathbb{1} \pm \Pi_z | \psi \rangle = \left(\mathbb{1} \pm \langle \Pi_z \rangle_\psi \right) dt \quad (1.43)$$

It is convenient to evaluate the direction of the unnormalized state $\hat{C}_\pm |++\rangle$:

$$(\sigma_z^{(1)} + \sigma_z^{(2)}) |++\rangle = 2(|11\rangle - |00\rangle) = -2\sqrt{2} |\phi_-\rangle, \quad (1.44)$$

$$(\sigma_z^{(1)} - \sigma_z^{(2)}) |++\rangle = 2(|10\rangle - |01\rangle) = -2\sqrt{2} |\psi_-\rangle. \quad (1.45)$$

Hence the *normalized* post-jump states are

$$|++\rangle \xrightarrow{C_+} -|\phi_-\rangle, \quad |++\rangle \xrightarrow{C_-} -|\psi_-\rangle. \quad (1.46)$$

Overall minus signs are global phases and carry no physical effect; we keep them to make the subsequent alternation visually clear.

Equations (1.46) show that the first click already fixes the z -parity: $\hat{C}_+$ lands in the even sector (Bell- ϕ states), while $\hat{C}_-$ lands in the odd sector (Bell- ψ states).

For the state $|++\rangle$, since $\langle \sigma_z \rangle_{|++\rangle} = 0$, we have $\langle \Pi_z \rangle_{|++\rangle} = \langle \sigma_z \rangle_{|+\rangle} \langle \sigma_z \rangle_{|+\rangle} = 0$, hence

$$p_+(|++\rangle) = p_-(|++\rangle) = \gamma dt. \quad (1.47)$$

The two channels are equiprobable on the initial state, and the total jump rate is $2\gamma dt$.

For any even-parity state (eigenvalue +1 of Π_z ; in particular $|\phi_\pm\rangle$),

$$p_+(\text{even}) = 2\gamma dt, \quad p_-(\text{even}) = 0. \quad (1.48)$$

For any odd-parity state (eigenvalue -1; in particular $|\psi_\pm\rangle$),

$$p_-(\text{odd}) = 2\gamma dt, \quad p_+(\text{odd}) = 0. \quad (1.49)$$

These two outcomes already show the key phenomenon: a single detected jump projects $|++\rangle$ onto a *definite* parity sector (even for $\hat{C}_+$, odd for $\hat{C}_-$), which we now formalize.

1.3.3 Subsequent jumps: two-state cyclicity

It follows that the action of the Lindbladians on the Bell states is:

$$\hat{C}_+ |\phi_-\rangle = -|\phi_+\rangle, \quad \hat{C}_+ |\phi_+\rangle = -|\phi_-\rangle, \quad (1.50)$$

$$\hat{C}_- |\psi_-\rangle = -|\psi_+\rangle, \quad \hat{C}_- |\psi_+\rangle = -|\psi_-\rangle. \quad (1.51)$$

After renormalization, this yields the deterministic cycles

$$-|\phi_-\rangle \xrightarrow{C_+} |\phi_+\rangle \xrightarrow{C_+} -|\phi_-\rangle \xrightarrow{C_+} \dots, \quad -|\psi_-\rangle \xrightarrow{C_-} |\psi_+\rangle \xrightarrow{C_-} -|\psi_-\rangle \xrightarrow{C_-} \dots. \quad (1.52)$$

Because of $[C_\pm, \Pi_z] = 0$, the “wrong” channel is forbidden inside each parity subspace.

1.3.4 Dynamics of CSS states

We now specialize the discussion to two-qubit spin coherent states (CSS) (1.4) and describe how the photodetection dynamics acts on them. We set $\phi = 0$.

By applying the Lindbladian operators (1.12), one obtains

$$\hat{C}_+ |\text{CSS}, \theta\rangle = \sin^2\left(\frac{\theta}{2}\right) |11\rangle - \cos^2\left(\frac{\theta}{2}\right) |00\rangle, \quad (1.53)$$

$$\hat{C}_- |\text{CSS}, \theta\rangle = \sin \frac{\theta}{2} \cos \frac{\theta}{2} (|10\rangle - |01\rangle). \quad (1.54)$$

After normalization, the conditional post-click pure states are

$$|\tilde{\phi}_\theta\rangle = \frac{\sin^2(\frac{\theta}{2}) |11\rangle - \cos^2(\frac{\theta}{2}) |00\rangle}{\sqrt{\sin^4(\frac{\theta}{2}) + \cos^4(\frac{\theta}{2})}}, \quad |\psi_-\rangle = \frac{|10\rangle - |01\rangle}{\sqrt{2}}. \quad (1.55)$$

The “−” click therefore projects onto a maximally entangled Bell state independently of θ , while the “+” click yields an even-parity state whose entanglement depends on θ .

After the first click, parity is fixed. Indeed,

$$\left(|10\rangle\langle 10| - |01\rangle\langle 01| \right) |\tilde{\phi}_\theta\rangle = 0, \quad \left(|11\rangle\langle 11| - |00\rangle\langle 00| \right) |\psi_-\rangle = 0. \quad (1.56)$$

Within each parity sector the jump action induces a two-state cyclicity. In the even sector we define

$$|\tilde{\phi}_{\pm, \theta}\rangle = \frac{\sin^2(\frac{\theta}{2}) |11\rangle \pm \cos^2(\frac{\theta}{2}) |00\rangle}{\sqrt{\sin^4(\frac{\theta}{2}) + \cos^4(\frac{\theta}{2})}}, \quad (1.57)$$

so it follows that we retrieve the more general deterministic cycles

$$|\text{CSS}, \theta\rangle \xrightarrow{\hat{C}_+} |\tilde{\phi}_{-, \theta}\rangle \xrightarrow{\hat{C}_+} |\tilde{\phi}_{+, \theta}\rangle \xrightarrow{\hat{C}_+} |\tilde{\psi}_{-, \theta}\rangle \quad |\text{CSS}, \theta\rangle \xrightarrow{\hat{C}_-} |\psi_-\rangle \xrightarrow{\hat{C}_-} |\psi_+\rangle \xrightarrow{\hat{C}_-} |\psi_-\rangle \quad (1.58)$$

1.3.5 Analyzing evolution and jumps: even-parity channel

Now we want to prove that evolving $|++\rangle$ with liouvillian as (1.10) is totally equivalent in evolving in time the first click state for both channel. We show it explicitly for the even-parity case, meaning:

$$\hat{C}_+ \cdot e^{\mathcal{L}_{(1-\eta)\gamma}^{(2)} t} [|++\rangle\langle ++|] \cdot \hat{C}_+ = e^{\mathcal{L}_{(1-\eta)\gamma}^{(2)} t} [|\psi_-\rangle\langle \psi_-|], \quad (1.59)$$

where $\mathcal{L}_{(1-\eta)\gamma}^{(2)}$ is the liouvillian in the ME.

Using the collective jump operators $\hat{C}_\pm$, we write

$$\mathcal{L}_{(1-\eta)\gamma}^{(2)} [\rho_t] = (1-\eta) \frac{\gamma}{2} \left(\mathcal{D}[\hat{C}_+] \rho_t + \mathcal{D}[\hat{C}_-] \rho_t \right). \quad (1.60)$$

Since the unmonitored dephasing channel acts locally on each qubit, for product inputs the evolution factorizes. In particular,

$$e^{\mathcal{L}_{(1-\eta)\gamma}^{(2)} t} [|++\rangle\langle ++|] = \left(e^{\mathcal{L}_{(1-\eta)\gamma}^{(1)} t} [|+\rangle\langle +|] \right) \otimes \left(e^{\mathcal{L}_{(1-\eta)\gamma}^{(1)} t} [|+\rangle\langle +|] \right), \quad (1.61)$$

where $\mathcal{L}_{(1-\eta)\gamma}^{(1)}$ denotes the corresponding single-qubit Liouvillian, the “single-qubit dephasing” channel.

We vectorize the single-qubit problem in the computational basis $\{|0\rangle, |1\rangle\}$. Let $\rho(t)$ be a single-qubit density operator written as

$$\rho(t) = \begin{pmatrix} \alpha_{00}(t) & \alpha_{01}(t) \\ \alpha_{10}(t) & \alpha_{11}(t) \end{pmatrix}, \quad (1.62)$$

and its equation of motion

$$\dot{\rho}(t) = \mathcal{L}_{(1-\eta)\gamma}^{(1)}[\rho(t)]. \quad (1.63)$$

In the computational basis, the superoperator $\mathcal{L}_{(1-\eta)\gamma}^{(1)}[\rho(t)]$ can be represented through its matrix elements as

$$\mathcal{L}_{(1-\eta)\gamma}^{(1)}[\rho(t)] \equiv \begin{pmatrix} \langle 0 | \mathcal{L}_{(1-\eta)\gamma}^{(1)}[\rho(t)] | 0 \rangle & \langle 0 | \mathcal{L}_{(1-\eta)\gamma}^{(1)}[\rho(t)] | 1 \rangle \\ \langle 1 | \mathcal{L}_{(1-\eta)\gamma}^{(1)}[\rho(t)] | 0 \rangle & \langle 1 | \mathcal{L}_{(1-\eta)\gamma}^{(1)}[\rho(t)] | 1 \rangle \end{pmatrix}. \quad (1.64)$$

Hence, to obtain the explicit differential equations for the coefficients $\alpha_{00}(t), \alpha_{01}(t), \alpha_{10}(t)$, one computes $\mathcal{L}_{(1-\eta)\gamma}^{(1)}[\rho(t)]$ and projects it onto the computational basis.

We consider the single-qubit dephasing generator (unmonitored part)

$$\dot{\rho}(t) = \mathcal{L}_{\Gamma}^{(1)}[\rho(t)] = \frac{\Gamma}{2} \left(\sigma_z \rho(t) \sigma_z - \rho(t) \right), \quad (1.65)$$

Then

$$\sigma_z \rho(t) \sigma_z = \begin{pmatrix} \alpha_{00}(t) & -\alpha_{01}(t) \\ -\alpha_{10}(t) & \alpha_{11}(t) \end{pmatrix}, \quad (1.66)$$

and therefore

$$\mathcal{L}_{(1-\eta)\gamma}^{(1)}[\rho(t)] = \frac{(1-\eta)\gamma}{2} \begin{pmatrix} 0 & -2\alpha_{01}(t) \\ -2\alpha_{10}(t) & 0 \end{pmatrix}. \quad (1.67)$$

Hence the populations are constants, while coherences decay exponentially:

$$\dot{\alpha}_{00}(t) = \dot{\alpha}_{11}(t) = 0, \quad (1.68)$$

$$\dot{\alpha}_{01}(t) = \dot{\alpha}_{10}(t) = -(1-\eta)\gamma \quad (1.69)$$

so that

$$\rho(t) = \begin{pmatrix} \alpha_{00}(0) & \alpha_{01}(0)e^{-(1-\eta)\gamma t} \\ \alpha_{10}(0)e^{-(1-\eta)\gamma t} & \alpha_{11}(0) \end{pmatrix}. \quad (1.70)$$

In particular, for $\rho(0) = |+\rangle\langle+|$ one has $\alpha_{jk} = 1/2 \forall j, k$, hence

$$\rho_+(t) = \begin{pmatrix} \frac{1}{2} & \frac{1}{2}e^{-(1-\eta)\gamma t} \\ \frac{1}{2}e^{-(1-\eta)\gamma t} & \frac{1}{2} \end{pmatrix}, \quad \rho_+(t \rightarrow \infty) = \frac{\mathbb{1}_2}{2}. \quad (1.71)$$

Since the channel acts independently on the two qubits,

$$\rho_{++}(t) = e^{\mathcal{L}_{(1-\eta)\gamma}^{(2)} t} [|++\rangle\langle++|] = \rho_+(t) \otimes \rho_+(t), \quad (1.72)$$

where $\rho_+(t)$ is given in Eq. (1.70). In the computational basis ordered as $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, this yields

$$\rho_{++}(t) = \frac{1}{4} \begin{pmatrix} 1 & e^{-(1-\eta)\gamma t} & e^{-(1-\eta)\gamma t} & e^{-2(1-\eta)\gamma t} \\ e^{-(1-\eta)\gamma t} & 1 & e^{-2(1-\eta)\gamma t} & e^{-(1-\eta)\gamma t} \\ e^{-(1-\eta)\gamma t} & e^{-2(1-\eta)\gamma t} & 1 & e^{-(1-\eta)\gamma t} \\ e^{-2(1-\eta)\gamma t} & e^{-(1-\eta)\gamma t} & e^{-(1-\eta)\gamma t} & 1 \end{pmatrix}. \quad (1.73)$$

We now apply the collective operator $\hat{C}_+ = \text{diag}(-\sqrt{2}, 0, 0, \sqrt{2})$, used to model the + jump channel. Only the $\{|00\rangle, |11\rangle\}$ block survives, and one finds

$$C_+ \rho_{++}(t) C_+^\dagger = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & -e^{-2(1-\eta)\gamma t} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -e^{-2(1-\eta)\gamma t} & 0 & 0 & 1 \end{pmatrix} \quad (1.74)$$

which is supported on the even subspace.

At $t = 0$,

$$C_+ \rho_{++}(0) C_+^\dagger = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix} = |\phi_-\rangle\langle\phi_-|. \quad (1.75)$$

We now compute the right-hand side

$$e^{\mathcal{L}_\gamma^{(2)} t} [|\phi_-\rangle\langle\phi_-|], \quad (1.76)$$

and show that it matches the expression obtained for the LHS.

The corresponding Liouvillian act as:

$$\dot{\rho}(t) = \mathcal{L}_\gamma^{(2)}[\rho(t)] = \frac{\gamma}{2} \left(\mathcal{D}[\hat{C}_+]\rho(t) + \mathcal{D}[\hat{C}_-]\rho(t) \right), \quad (1.77)$$

Since $|\phi_-\rangle$ has even z -parity, one has $\hat{C}_- |\phi_-\rangle = 0$, hence

$$\mathcal{D}[\hat{C}_-](|\phi_-\rangle\langle\phi_-|) = 0, \quad (1.78)$$

and the dynamics reduces to the + channel.

Using (1.37) the dissipator can be rewritten as

$$\begin{aligned} \mathcal{D}[\hat{C}_\pm]\rho &= \hat{C}_\pm \rho \hat{C}_\pm - \frac{1}{2} \{(\mathbb{1} \pm \Pi_z), \rho\} \\ &= \hat{C}_\pm \rho \hat{C}_\pm - \rho \mp \frac{1}{2} (\Pi_z \rho + \rho \Pi_z). \end{aligned} \quad (1.79)$$

For parity-preserving states (block-diagonal with respect to even/odd parity) one has $[\rho, \Pi_z] = 0$. In the even subspace $\text{span}\{|00\rangle, |11\rangle\}$, $\Pi_z = +\mathbb{1}$, hence $\Pi_z \rho + \rho \Pi_z = 2\rho$ and therefore

$$\mathcal{D}[\hat{C}_+]\rho = \hat{C}_+ \rho \hat{C}_+ - 2\rho \quad (\text{even subspace}). \quad (1.80)$$

Restricting to the even basis $\{|00\rangle, |11\rangle\}$, define the reduced 2×2 state

$$\tilde{\rho}(t) = \begin{pmatrix} \alpha_{00,00}(t) & \alpha_{00,11}(t) \\ \alpha_{11,00}(t) & \alpha_{11,11}(t) \end{pmatrix}. \quad (1.81)$$

In this basis,

$$\tilde{C}_+ = \frac{1}{\sqrt{2}} \begin{pmatrix} -2 & 0 \\ 0 & +2 \end{pmatrix} = \sqrt{2} \tilde{\sigma}_z, \quad \tilde{\sigma}_z = \begin{pmatrix} -1 & 0 \\ 0 & +1 \end{pmatrix}. \quad (1.82)$$

Using Eq. (1.80), the reduced equation of motion becomes

$$\begin{aligned} \dot{\tilde{\rho}}(t) &= \frac{\gamma}{2} \left(\tilde{C}_+ \tilde{\rho}(t) \tilde{C}_+ - 2 \tilde{\rho}(t) \right) \\ &= \frac{\gamma}{2} \left(2 \tilde{\sigma}_z \tilde{\rho}(t) \tilde{\sigma}_z - 2 \tilde{\rho}(t) \right) = \gamma \left(\tilde{\sigma}_z \tilde{\rho}(t) \tilde{\sigma}_z - \tilde{\rho}(t) \right). \end{aligned} \quad (1.83)$$

Explicitly,

$$\tilde{\sigma}_z \tilde{\rho}(t) \tilde{\sigma}_z = \begin{pmatrix} \alpha_{00,00}(t) & -\alpha_{00,11}(t) \\ -\alpha_{11,00}(t) & \alpha_{11,11}(t) \end{pmatrix}, \quad (1.84)$$

so that

$$\dot{\tilde{\rho}}(t) = \gamma \begin{pmatrix} 0 & -2\alpha_{00,11}(t) \\ -2\alpha_{11,00}(t) & 0 \end{pmatrix}. \quad (1.85)$$

Hence the populations are constants and the coherences decay as $e^{-2\gamma t}$:

$$\tilde{\rho}(t) = \begin{pmatrix} \alpha_{00,00}(0) & \alpha_{00,11}(0)e^{-2\gamma t} \\ \alpha_{11,00}(0)e^{-2\gamma t} & \alpha_{11,11}(0) \end{pmatrix}. \quad (1.86)$$

For $\tilde{\rho}(0) = |\phi_-\rangle\langle\phi_-|$ one has

$$\tilde{\rho}(0) = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad (1.87)$$

thus

$$\tilde{\rho}(t) = \frac{1}{2} \begin{pmatrix} 1 & -e^{-2\gamma t} \\ -e^{-2\gamma t} & 1 \end{pmatrix}. \quad (1.88)$$

Embedding back into the full computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ yields

$$e^{\mathcal{L}_\gamma^{(2)} t} [|\phi_-\rangle\langle\phi_-|] = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & -e^{-2\gamma t} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -e^{-2\gamma t} & 0 & 0 & 1 \end{pmatrix} \quad (1.89)$$

which matches the LHS expression upon the replacement $\gamma \mapsto (1-\eta)\gamma$ when interpreting $\mathcal{L}^{(2)}$ as the unmonitored part of the inefficient photodetection dynamics.

1.3.6 Analyzing evolution and jumps: odd-parity channel case

We now repeat the same computation for the negative channel. Explicitly,

$$\hat{C}_- \cdot e^{\mathcal{L}_{(1-\eta)\gamma}^{(2)} t} [|++\rangle\langle++|] \cdot \hat{C}_-^\dagger \quad (1.90)$$

The unmonitored generator can be written in terms of collective operators as

$$\mathcal{L}_{(1-\eta)\gamma}^{(2)}[\rho_t] = \frac{(1-\eta)\gamma}{2} \left(\mathcal{D}[\hat{C}_+] \rho_t + \mathcal{D}[\hat{C}_-] \rho_t \right), \quad (1.91)$$

By $\hat{C}_- = \text{diag}(0, -\sqrt{2}, \sqrt{2}, 0)$, one obtains

$$\hat{C}_- \rho_{++}(t) \hat{C}_-^\dagger = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -e^{-2(1-\eta)\gamma t} & 0 \\ 0 & -e^{-2(1-\eta)\gamma t} & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (1.92)$$

which is supported on the odd subspace $\text{span}\{|01\rangle, |10\rangle\}$.

At $t = 0$,

$$\hat{C}_- \rho_{++}(0) \hat{C}_-^\dagger = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = |\psi_-\rangle \langle \psi_-|. \quad (1.93)$$

We now compute the evolution starting from $|\psi_-\rangle$ under the same Liouvillian and show that it reproduces Eq. (1.92).

Since $|\psi_-\rangle$ has odd z -parity, one has $\hat{C}_- |\psi_-\rangle = 0$, hence

$$\mathcal{D}[\hat{C}_+](|\psi_-\rangle \langle \psi_-|) = 0, \quad (1.94)$$

and the dynamics reduces to the minus channel. Restricting to the odd basis $\{|01\rangle, |10\rangle\}$, define the reduced 2×2 state

$$\tilde{\rho}(t) = \begin{pmatrix} \alpha_{01,01}(t) & \alpha_{01,10}(t) \\ \alpha_{10,01}(t) & \alpha_{10,10}(t) \end{pmatrix}. \quad (1.95)$$

For odd-parity states, the identity $\hat{C}_-^\dagger \hat{C}_- = \mathbb{1} - \Pi_z$ reduces to $\hat{C}_-^\dagger \hat{C}_- = 2 \cdot \mathbb{1}$ on the odd subspace. Therefore,

$$\mathcal{D}[\hat{C}_-] \rho = \hat{C}_- \rho \hat{C}_- - 2\rho \quad (\text{odd subspace}). \quad (1.96)$$

The reduced equation of motion becomes

$$\dot{\tilde{\rho}}(t) = (1-\eta)\gamma \left(\tilde{\sigma}_z \tilde{\rho}(t) \tilde{\sigma}_z - \tilde{\rho}(t) \right). \quad (1.97)$$

Explicitly,

$$\dot{\tilde{\rho}}(t) = (1-\eta)\gamma \begin{pmatrix} 0 & -2\alpha_{01,10}(t) \\ -2\alpha_{10,01}(t) & 0 \end{pmatrix}. \quad (1.98)$$

Hence the populations are constants and the coherences decay as $e^{-2(1-\eta)\gamma t}$:

$$\tilde{\rho}(t) = \begin{pmatrix} \alpha_{01,01}(0) & \alpha_{01,10}(0)e^{-2(1-\eta)\gamma t} \\ \alpha_{10,01}(0)e^{-2(1-\eta)\gamma t} & \alpha_{10,10}(0) \end{pmatrix}. \quad (1.99)$$

For $\tilde{\rho}(0) = |\psi_-\rangle\langle\psi_-|$ one has

$$\tilde{\rho}(t) = \frac{1}{2} \begin{pmatrix} 1 & -e^{-2(1-\eta)\gamma t} \\ -e^{-2(1-\eta)\gamma t} & 1 \end{pmatrix}. \quad (1.100)$$

Embedding back into the full computational basis yields

$$e^{\mathcal{L}_{(1-\eta)\gamma}^{(2)} t} [|\psi_-\rangle\langle\psi_-|] = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -e^{-2(1-\eta)\gamma t} & 0 \\ 0 & -e^{-2(1-\eta)\gamma t} & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (1.101)$$

which matches Eq. (1.92). Hence Eq. (1.90) is proven.

1.3.7 Ensemble-averaged quantities over trajectories

We now compute expectation values over the photodetection trajectories. Let $P_0(t)$ denote the probability of observing no click up to time t . Since total rate is 2γ , the waiting-time distribution is exponential and

$$P_0(t) = e^{-2\gamma t}, \quad P_{\text{click}}(t) = 1 - P_0(t) = 1 - e^{-2\gamma t}. \quad (1.102)$$

The ensemble-averaged concurrence can be written as

$$\mathbb{E}[\mathcal{C}(t)] = \mathbb{E}[\mathcal{C}(t) | \text{no-click}] P_0(t) + \mathbb{E}[\mathcal{C}(t) | \text{click}] P_{\text{click}}(t). \quad (1.103)$$

For the present dynamics, $\mathcal{C} = 0$ along the no-click branch (the state remains separable), while $\mathcal{C} = 1$ after the first click (the state is projected onto a Bell state). Hence

$$\mathbb{E}[\mathcal{C}(t)] = 1 - e^{-2\gamma t}. \quad (1.104)$$

We proceed similarly for the variance $\Delta J_z^2(t)$. Splitting the trajectories into the no-click part and the click part, one has

$$\mathbb{E}[\Delta J_z^2(t)] = \mathbb{E}[\Delta J_z^2(t) | \text{no-click}] P_0(t) + \mathbb{E}[\Delta J_z^2(t) | \text{click}] P_{\text{click}}(t). \quad (1.105)$$

For an initial CSS state we see that the weights of even/odd sectors are

$$P_+(\theta) = \sin^4\left(\frac{\theta}{2}\right) + \cos^4\left(\frac{\theta}{2}\right), \quad P_-(\theta) = 2 \sin^2\left(\frac{\theta}{2}\right) \cos^2\left(\frac{\theta}{2}\right), \quad P_+ + P_- = 1. \quad (1.106)$$

Along the no-click branch the state stays equal (up to normalization) to the initial CSS, hence

$$\Delta J_z^2(\text{no-click}) = \frac{\sin^2 \theta}{2}. \quad (1.107)$$

After the first click, the state is projected onto the even-parity Bell sector with weight P_+ or onto the odd-parity Bell sector with weight P_- ; using

$$\Delta J_z^2(|\phi_{\pm}\rangle) = 1, \quad \Delta J_z^2(|\psi_{\pm}\rangle) = 0, \quad (1.108)$$

one obtains

$$\mathbb{E}[\Delta J_z^2(t) | \text{click}] = P_+(\theta) \cdot 1 + P_-(\theta) \cdot 0 = P_+(\theta). \quad (1.109)$$

Therefore

$$\mathbb{E}[\Delta J_z^2(t)] = e^{-2\gamma t} \frac{\sin^2 \theta}{2} + (1 - e^{-2\gamma t}) \left[\sin^4\left(\frac{\theta}{2}\right) + \cos^4\left(\frac{\theta}{2}\right) \right]. \quad (1.110)$$

1.3.8 Spin-squeezing

We want to evaluate the spin-squeezing parameter. We start from the mean spin direction (MSD) of the post-click states generated from $|\text{CSS}, \theta\rangle$. We define the collective spin operators (for $N = 2$)

$$\hat{J}_k \equiv \frac{1}{2} \left(\sigma_i^{(1)} + \sigma_i^{(2)} \right), \quad i \in \{x, y, z\}, \quad (1.111)$$

and the mean spin direction as

$$\mathbf{n}_0 \equiv \frac{\langle \hat{\mathbf{J}} \rangle}{|\langle \hat{\mathbf{J}} \rangle|}, \quad \langle \hat{\mathbf{J}} \rangle \equiv \left(\langle \hat{J}_x \rangle, \langle \hat{J}_y \rangle, \langle \hat{J}_z \rangle \right). \quad (1.112)$$

For $N = 2$, the Kitagawa–Ueda parameter is

$$\xi_{KU}^2 \equiv \frac{4}{N} \min_{\mathbf{n}_\perp} (\Delta \hat{J}_{\mathbf{n}_\perp})^2 = 2 \min_{\mathbf{n}_\perp} (\Delta \hat{J}_{\mathbf{n}_\perp})^2, \quad \hat{J}_{\mathbf{n}} \equiv \mathbf{n} \cdot \hat{\mathbf{J}}, \quad (1.113)$$

where the minimization is taken over directions orthogonal to the mean spin direction (MSD). If the MSD is not defined, i.e. $\langle \hat{\mathbf{J}} \rangle = \mathbf{0}$, we adopt the natural extension given by minimizing over all directions $\mathbf{n}$.

We start from the trajectory decomposition. Along the no-click branch, the normalized conditional state remains equal to the initial CSS, hence

$$\xi_{KU}^2(\text{no-click}) = 1. \quad (1.114)$$

This follows from the defining property of CSS states, for which the minimal transverse variance equals $N/4$, i.e. $\min_{\mathbf{n}_\perp} (\Delta \hat{J}_{\mathbf{n}_\perp})^2 = N/4 = 1/2$ when $N = 2$.

We now compute ξ_{KU}^2 for the post-click states generated from $|\text{CSS}, \theta\rangle$. We use the notation introduced in Eq. (1.58):

$$\left| \tilde{\psi}_{\pm, \theta} \right\rangle = \frac{\sin^2(\frac{\theta}{2}) |11\rangle \pm \cos^2(\frac{\theta}{2}) |00\rangle}{\sqrt{\sin^4(\frac{\theta}{2}) + \cos^4(\frac{\theta}{2})}}, \quad |\psi_{\pm}\rangle = \frac{|10\rangle \pm |01\rangle}{\sqrt{2}}. \quad (1.115)$$

We now consider the odd-parity Bell states, related to the $\hat{C}_-$ channel. Since every Bell state has vanishing expectation value for all spin components, $\mathbf{n}_0$ is not defined for any post-click state. Therefore we have to minimize the variance all over the directions, and it is known that it is 0. Hence the odd-parity click channel yields $\xi_{KU}^2 = 0$, independently of θ , and subsequent clicks do not change this value.

We start from the even-parity post-click family $\left| \tilde{\psi}_{\pm, \theta} \right\rangle$ as defined in (1.57). First, we compute $\langle \hat{J}_z \rangle$, by setting

$$A \equiv \sin^2\left(\frac{\theta}{2}\right), \quad B \equiv \cos^2\left(\frac{\theta}{2}\right), \quad \mathcal{N} \equiv A^2 + B^2, \quad (1.116)$$

we obtain

$$\langle \hat{J}_z \rangle_{\tilde{\psi}_{\pm, \theta}} = \langle \tilde{\psi}_{\pm, \theta} | \hat{J}_z | \tilde{\psi}_{\pm, \theta} \rangle = \frac{1}{\mathcal{N}} \left(A^2 \langle 11 | \hat{J}_z | 11 \rangle + B^2 \langle 00 | \hat{J}_z | 00 \rangle \right) = \frac{A^2 - B^2}{A^2 + B^2}. \quad (1.117)$$

Since $A^2 - B^2 = \sin^4(\theta/2) - \cos^4(\theta/2) = \sin^2(\theta/2) - \cos^2(\theta/2) = -\cos \theta$, we can rewrite Eq. (1.117) as

$$\boxed{\langle \hat{J}_z \rangle_{\tilde{\psi}_{\pm, \theta}} = \frac{-\cos \theta}{\sin^4(\frac{\theta}{2}) + \cos^4(\frac{\theta}{2})}.} \quad (1.118)$$

We now compute $\langle \hat{J}_x \rangle$ and $\langle \hat{J}_y \rangle$. They both map the even subspace $\text{span}\{|00\rangle, |11\rangle\}$ into the odd subspace $\text{span}\{|01\rangle, |10\rangle\}$ and viceversa. Since our state has support only on the even sector, both expectation values vanish. Hence the MSD is along the z axis whenever $\cos \theta \neq 0$:

$$\boxed{\mathbf{n}_0(\tilde{\psi}_{\pm, \theta}) = \text{sgn}(-\cos \theta) \hat{\mathbf{z}}, \quad \text{for } \theta \neq \frac{\pi}{2}.} \quad (1.119)$$

At $\theta = \pi/2$ one has $\langle \hat{J}_z \rangle = 0$ and therefore $|\langle \hat{\mathbf{J}} \rangle| = 0$, so $\mathbf{n}_0$ is not defined. In the xy plane we introduce the rotated operator

$$\hat{J}_\alpha = \cos \alpha \hat{J}_x + \sin \alpha \hat{J}_y, \quad \alpha \in [0, 2\pi). \quad (1.120)$$

Its variance can be written in terms of the 2×2 covariance matrix in the (x, y) sector. Explicitly,

$$(\Delta J_\alpha)^2 = \cos^2 \alpha (\Delta J_x)^2 + \sin^2 \alpha (\Delta J_y)^2 + \sin \alpha \cos \alpha \left(\frac{1}{2} \langle \{\hat{J}_x, \hat{J}_y\} \rangle - \langle \hat{J}_x \rangle \langle \hat{J}_y \rangle - \langle \hat{J}_y \rangle \langle \hat{J}_x \rangle \right), \quad (1.121)$$

that is

$$(\Delta J_\alpha)^2 = \cos^2 \alpha (\Delta J_x)^2 + \sin^2 \alpha (\Delta J_y)^2 + \sin \alpha \cos \alpha \left(\frac{1}{2} \langle \{\hat{J}_x, \hat{J}_y\} \rangle - 2 \langle \hat{J}_x \rangle \langle \hat{J}_y \rangle \right). \quad (1.122)$$

In general, the minimization over α corresponds to taking the smallest eigenvalue of the transverse covariance matrix. By parity considerations one has $\langle \hat{J}_x \rangle = \langle \hat{J}_y \rangle = 0$ on each post-click branch. Hence Eq. (1.122) reduces to

$$(\Delta J_\alpha)^2 = \cos^2 \alpha (\Delta J_x)^2 + \sin^2 \alpha (\Delta J_y)^2. \quad (1.123)$$

The minimum is then attained at one of the axes:

$$\min_{\alpha} (\Delta J_\alpha)^2 = \min\{(\Delta J_x)^2, (\Delta J_y)^2\}. \quad (1.124)$$

Therefore, for the present post-click states it is sufficient to compare $(\Delta J_x)^2$ and $(\Delta J_y)^2$, without scanning all transverse directions.

We use the identities

$$\hat{J}_x^2 = \frac{1}{4} \left[(\sigma_x^{(1)})^2 + (\sigma_x^{(2)})^2 + 2\sigma_x^{(1)}\sigma_x^{(2)} \right] = \frac{1}{2} \left(\mathbb{1} + \sigma_x^{(1)}\sigma_x^{(2)} \right), \quad (1.125)$$

and similarly

$$\hat{J}_y^2 = \frac{1}{2} \left(\mathbb{1} + \sigma_y^{(1)}\sigma_y^{(2)} \right). \quad (1.126)$$

Since $|\tilde{\psi}_{\pm,\theta}\rangle$ is supported on $\text{span}\{|00\rangle, |11\rangle\}$, we only need the action of $\sigma_x^{(1)}\sigma_x^{(2)}$ and $\sigma_y^{(1)}\sigma_y^{(2)}$ on that subspace. One has

$$\sigma_x^{(1)}\sigma_x^{(2)}|00\rangle = |11\rangle, \quad \sigma_x^{(1)}\sigma_x^{(2)}|11\rangle = |00\rangle, \quad (1.127)$$

and

$$\sigma_y^{(1)}\sigma_y^{(2)}|00\rangle = -|11\rangle, \quad \sigma_y^{(1)}\sigma_y^{(2)}|11\rangle = -|00\rangle. \quad (1.128)$$

Hence

$$\langle \sigma_x^{(1)}\sigma_x^{(2)} \rangle_{\tilde{\psi}_{\pm,\theta}} = \frac{\pm 2AB}{P_+}, \quad \langle \sigma_y^{(1)}\sigma_y^{(2)} \rangle_{\tilde{\psi}_{\pm,\theta}} = \frac{\mp 2AB}{P_+}. \quad (1.129)$$

Since $\langle \hat{J}_x \rangle = \langle \hat{J}_y \rangle = 0$, we obtain

$$\Delta \hat{J}_x^2(\tilde{\psi}_{\pm,\theta}) = \langle \hat{J}_x^2 \rangle = \frac{1}{2} \left(1 \pm \frac{2AB}{P_+} \right) = \frac{(A \pm B)^2}{2P_+}, \quad (1.130)$$

and

$$\Delta \hat{J}_y^2(\tilde{\psi}_{\pm,\theta}) = \langle \hat{J}_y^2 \rangle = \frac{1}{2} \left(1 \mp \frac{2AB}{P_+} \right) = \frac{(A \mp B)^2}{2P_+}. \quad (1.131)$$

Therefore the minimal transverse variance is the same for both $|\tilde{\psi}_{-,\theta}\rangle$ and $|\tilde{\psi}_{+,\theta}\rangle$, because it is always given by the smaller between $(A+B)^2$ and $(A-B)^2$. Since $A, B \geq 0$, one has $(A-B)^2 \leq (A+B)^2$, hence

$$\min_{\mathbf{n}_\perp} (\Delta \hat{J}_{\mathbf{n}_\perp})_{\tilde{\psi}_{\pm,\theta}}^2 = \frac{(A-B)^2}{2P_+}. \quad (1.132)$$

For $N = 2$ this yields

$$\xi_{KU}^2(\tilde{\psi}_{\pm,\theta}) = 2 \cdot \frac{(A-B)^2}{2P_+} = \frac{(A-B)^2}{P_+}. \quad (1.133)$$

Using $A-B = \sin^2(\theta/2) - \cos^2(\theta/2) = -\cos\theta$ and $P_+ = \sin^4(\theta/2) + \cos^4(\theta/2)$, we obtain the compact form

$$\xi_{KU}^2(\tilde{\psi}_{\pm,\theta}) = \frac{\cos^2\theta}{\sin^4(\frac{\theta}{2}) + \cos^4(\frac{\theta}{2})}. \quad (1.134)$$

We now compute the ensemble-averaged spin squeezing, following the same splitting used previously. We write

$$\mathbb{E}[\xi_{KU}^2(t)] = \mathbb{E}[\xi_{KU}^2(t) | \text{no-click}] P_0(t) + \mathbb{E}[\xi_{KU}^2(t) | \text{click}] P_{\text{click}}(t), \quad P_{\text{click}}(t) = 1 - P_0(t). \quad (1.135)$$

In the present dynamics the no-click branch yields $\xi_{KU}^2 = 1$, cf. Eq. (1.114). Moreover, a click projects either onto the even sector with probability P_+ or onto the odd sector with probability P_- , where

$$P_+ = \sin^4\left(\frac{\theta}{2}\right) + \cos^4\left(\frac{\theta}{2}\right), \quad P_- = \frac{1}{2} \sin^2\theta, \quad P_+ + P_- = 1. \quad (1.136)$$

In the odd sector we have $\xi_{KU}^2 = 0$ while in the even sector we have Eq. (1.134). Therefore

$$\mathbb{E}[\xi_{KU}^2(t) | \text{click}] = P_+ \xi_{KU}^2(\tilde{\psi}_{\pm, \theta}) + P_- \cdot 0 = P_+ \frac{\cos^2 \theta}{P_+} = \cos^2 \theta. \quad (1.137)$$

Finally,

$$\boxed{\mathbb{E}[\xi_{KU}^2(t)] = P_0(t) \cdot 1 + (1 - P_0(t)) \cos^2 \theta.} \quad (1.138)$$

Using the same no-click probability as in the previous sections, $P_0(t) = e^{-2\gamma t}$, we obtain

$$\boxed{\mathbb{E}[\xi_{KU}^2(t)] = e^{-2\gamma t} + (1 - e^{-2\gamma t}) \cos^2 \theta.} \quad (1.139)$$

For the special case $\theta = \pi/2$, i.e. $|\text{CSS}, \theta\rangle = |++\rangle$, one has $\cos^2 \theta = 0$ and

$$\mathbb{E}[\xi_{KU}^2(t)] = e^{-2\gamma t}, \quad (1.140)$$

which decays to zero as $t \rightarrow \infty$, consistently with the fact that after the first click the state is confined to Bell states with vanishing minimal collective-spin variance.

1.4 Homodyne detection

We now consider a diffusive unravelling of the same Markovian dynamics, obtained via homodyne detection. In this scheme the environment is continuously monitored and the measurement record is a real-valued photocurrent. The conditional state evolves stochastically, yet its ensemble average reproduces the GKSL master equation.

We start from a generic set of collapse operators $\{\hat{C}_k\}$. We denote by $\rho^{(c)}(t)$ the normalized conditional state. We introduce independent Wiener increments $\{dW_k(t)\}$ such that

$$\mathbb{E}[dW_k(t)] = 0, \quad dW_k(t) dW_\ell(t) = \delta_{k\ell} dt. \quad (1.141)$$

The detector efficiency is $\eta \in [0, 1]$. The local-oscillator (LO) phase θ selects which field quadrature is measured.

For each monitored channel k , the infinitesimal homodyne record can be written as

$$dY_k(t) = \sqrt{\eta} \left\langle e^{-i\theta} \hat{C}_k + e^{i\theta} \hat{C}_k^\dagger \right\rangle_t dt + dW_k(t). \quad (1.142)$$

The innovation process is therefore $dW_k(t) = dY_k(t) - \sqrt{\eta} \left\langle e^{-i\theta} \hat{C}_k + e^{i\theta} \hat{C}_k^\dagger \right\rangle_t dt$.

The corresponding Itô stochastic master equation (SME) reads

$$d\rho^{(c)} = -i[\hat{H}, \rho^{(c)}] dt + \sum_k \mathcal{D}[\hat{C}_k] \rho^{(c)} dt + \sqrt{\eta} \sum_k \mathcal{H}[e^{i\theta} \hat{C}_k] \rho^{(c)} dW_k(t), \quad (1.143)$$

where

$$\mathcal{H}[\hat{A}] \rho \equiv \hat{A} \rho + \rho \hat{A}^\dagger - \text{Tr}[(\hat{A} + \hat{A}^\dagger) \rho] \rho. \quad (1.144)$$

By construction, the average $\mathbb{E}[d\rho^{(c)}]$ reproduces the unconditional GKSL equation.

We now specialize to our two-qubit dephasing model. We use the collective jump operators introduced in Eq. (1.12),

$$\hat{C}_{\pm} = \sqrt{\gamma} \frac{\sigma_z^{(1)} \pm \sigma_z^{(2)}}{\sqrt{2}}. \quad (1.145)$$

We monitor the two output channels associated with $\hat{C}_+$ and $\hat{C}_-$. The homodyne SME becomes

$$\begin{aligned} d\rho^{(c)} = & -i[\hat{H}, \rho^{(c)}] dt + \mathcal{D}[\hat{C}_+] \rho^{(c)} dt + \mathcal{D}[\hat{C}_-] \rho^{(c)} dt \\ & + \sqrt{\eta} \mathcal{H}[e^{i\theta} \hat{C}_+] \rho^{(c)} dW_+(t) + \sqrt{\eta} \mathcal{H}[e^{i\theta} \hat{C}_-] \rho^{(c)} dW_-(t), \end{aligned} \quad (1.146)$$

with measurement records

$$\begin{aligned} dY_+(t) &= \sqrt{\eta} \left\langle e^{-i\theta} \hat{C}_+ + e^{i\theta} \hat{C}_+^\dagger \right\rangle_t dt + dW_+(t), \\ dY_-(t) &= \sqrt{\eta} \left\langle e^{-i\theta} \hat{C}_- + e^{i\theta} \hat{C}_-^\dagger \right\rangle_t dt + dW_-(t). \end{aligned} \quad (1.147)$$

A crucial simplification occurs in our model because $\hat{C}_{\pm}$ are Hermitian: $\hat{C}_{\pm}^\dagger = \hat{C}_{\pm}$. Then Eq. (1.142) reduces to

$$dY_{\pm}(t) = 2\sqrt{\eta} \cos \theta \left\langle \hat{C}_{\pm} \right\rangle_t dt + dW_{\pm}(t). \quad (1.148)$$

Therefore, the amount of information in the record is controlled by $\cos \theta$.

If $\theta = 0$, the record depends on $\left\langle \hat{C}_{\pm} \right\rangle_t$. The SME is nonlinear through the innovation term $\mathcal{H}[\hat{C}_{\pm}] \rho^{(c)}$. This corresponds to an informational measurement of the collective observables $\hat{C}_{\pm}$.

If $\theta = \pi/2$, one has $\cos \theta = 0$ and hence

$$dY_{\pm}(t) = dW_{\pm}(t). \quad (1.149)$$

The record contains only noise and no information about the system state. In this case the innovation superoperator becomes a commutator. Indeed, for Hermitian $\hat{C}$ one has

$$\mathcal{H}[i\hat{C}] \rho = i[\hat{C}, \rho], \quad (1.150)$$

since $(i\hat{C})^\dagger = -i\hat{C}$ and $\text{Tr}[(i\hat{C} - i\hat{C})\rho] = 0$. Therefore, at $\theta = \pi/2$ Eq. (1.146) becomes

$$\begin{aligned} d\rho^{(c)} = & -i[\hat{H}, \rho^{(c)}] dt + \mathcal{D}[\hat{C}_+] \rho^{(c)} dt + \mathcal{D}[\hat{C}_-] \rho^{(c)} dt \\ & + i\sqrt{\eta} [\hat{C}_+, \rho^{(c)}] dW_+(t) + i\sqrt{\eta} [\hat{C}_-, \rho^{(c)}] dW_-(t). \end{aligned} \quad (1.151)$$

This is a random-unitary backaction driven by Wiener noise, with a state-independent measurement record. It is the diffusive analogue of the photodetection case with unitary jumps: monitoring does not reveal information, yet it induces stochastic phase kicks.

Equation (1.146) provides the starting point for the analysis of homodyne-based entangling unravellings in the two-qubit dephasing model, by selecting the monitored quadrature through θ and by exploiting collective channels $\hat{C}_{\pm}$.

In our model $\hat{C}_\pm^\dagger = \hat{C}_\pm$. Hence

$$\hat{A}_\pm + \hat{A}_\pm^\dagger = (e^{i\theta} + e^{-i\theta}) \hat{C}_\pm = 2 \cos \theta \hat{C}_\pm. \quad (1.152)$$

This yields to the following decomposition:

$$\boxed{\mathcal{H}[e^{i\theta}\hat{C}]\rho = \cos \theta \left(\{\hat{C}, \rho\} - 2 \langle \hat{C} \rangle \rho \right) + i \sin \theta [\hat{C}, \rho], \quad (\hat{C}^\dagger = \hat{C}).} \quad (1.153)$$

Two limits are particularly relevant.

If $\theta = 0$, one obtains the informational (nonlinear) term

$$\mathcal{H}[\hat{C}]\rho = \{\hat{C}, \rho\} - 2 \langle \hat{C} \rangle \rho. \quad (1.154)$$

If $\theta = \pi/2$, one obtains a purely commutator form

$$\mathcal{H}[i\hat{C}]\rho = i[\hat{C}, \rho], \quad (1.155)$$

which is linear in ρ .

If $[\rho, \Pi_z] = 0$ at some time, then

$$[\mathcal{D}[\hat{C}_\pm]\rho, \Pi_z] = 0, \quad [\mathcal{H}[e^{i\theta}\hat{C}_\pm]\rho, \Pi_z] = 0. \quad (1.156)$$

Hence the homodyne SME preserves the block structure in the even/odd parity sectors.

1.4.1 Pure-state unravelling: homodyne SSE

For $\eta = 1$, if the initial conditional state is pure, it remains pure. We set $\rho^{(c)} = |\psi\rangle\langle\psi|$ and write an Itô stochastic Schrödinger equation (SSE). For each channel k we define $\hat{A}_k = e^{i\theta}\hat{C}_k$ and

$$m_k \equiv \langle \hat{A}_k + \hat{A}_k^\dagger \rangle. \quad (1.157)$$

A normalized SSE reproducing Eq. (1.143) is

$$d|\psi\rangle = \left[-i\hat{H} - \frac{1}{2} \sum_k \hat{C}_k^\dagger \hat{C}_k + \frac{1}{2} \sum_k m_k \hat{A}_k - \frac{1}{8} \sum_k m_k^2 \right] |\psi\rangle dt + \sum_k \left(\hat{A}_k - \frac{m_k}{2} \right) |\psi\rangle dW_k(t). \quad (1.158)$$

For Hermitian $\hat{C}_k$ one has $m_k = 2 \cos \theta \langle \hat{C}_k \rangle$. In particular, at $\theta = \pi/2$ one has $m_k = 0$ and the stochastic term becomes proportional to $i\hat{C}_k$.

Equation (1.158) will be used to propagate pure initial states such as $|++\rangle$, and to connect the wavefunction picture to the density-operator SME via $d\rho^{(c)} = d|\psi\rangle\langle\psi| + |\psi\rangle d\langle\psi| + d|\psi\rangle d\langle\psi|$.